

Seeing the Invisible

Physics-Guided Missing-Physics Learning for PEMFC Flooding Prediction

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ABSTRACT

Liquid water accumulation in porous gas diffusion layers limits mass transport in polymer electrolyte membrane fuel cells, while uncertain constitutive relations remain a major source of prediction error. We propose a physics-informed grey-box framework that retains the governing equations in a finite-element solver and learns only the missing gas-phase transport closure. The closure is identified from synthetic reference responses using discrete-adjoint sensitivities without prescribing its functional form. After training at RH 50%, the learned closure is frozen and evaluated zero-shot at unseen RH 70% on 16×40 and 32×80 meshes. The recovered law reaches 2.61% relative L^2 error, while saturation-field errors remain below 0.56%. Field-level agreement is retained without retraining, supporting transfer across the tested humidity and discretization changes and establishing a foundation for physics-guided digital twins of PEMFC flooding.

INTRODUCTION

Why flooding matters



What limits flooding prediction?

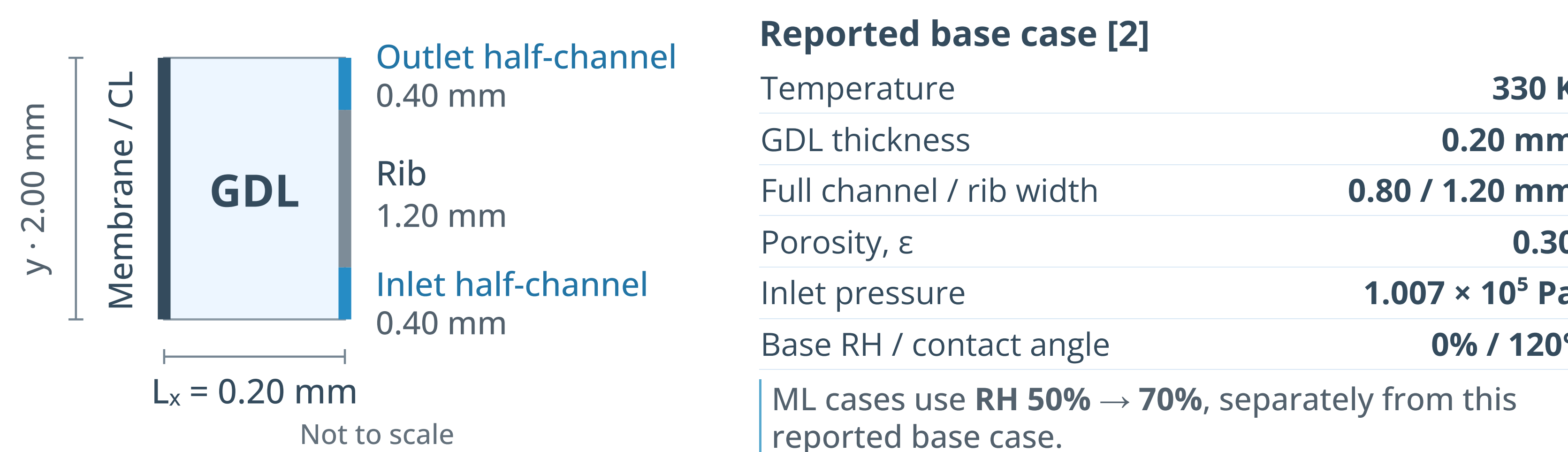
Observation	Internal liquid-water saturation $s(x,y)$ cannot be directly and continuously observed during operation.
Modeling	Flooding predictions depend on uncertain constitutive relations such as $D_{\text{eff}}(s)$, $k_r(s)$, and $p_c(s)$.

Existing ML vs our learning objective

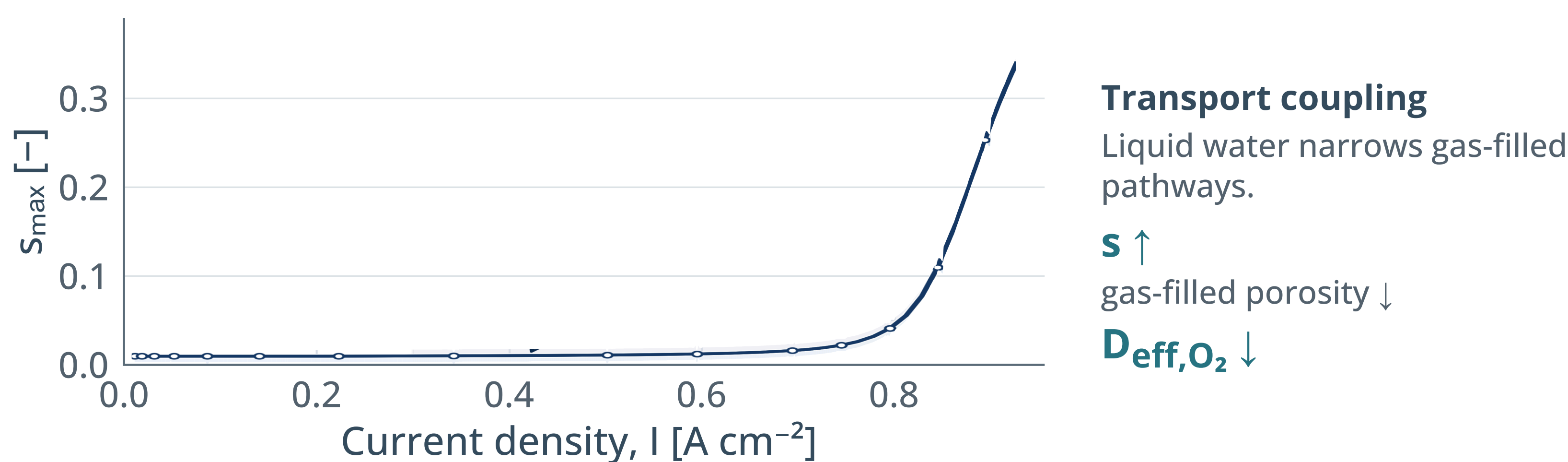
PINN Learn the PDE solution Physics is enforced through equation-residual training.	FNO / Neural operator Learn the solution operator Map input fields or parameters to solution fields.	OUR APPROACH FEM + Missing Physics ML Learn the constitutive law Only the uncertain transport relation is learned.
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RESULTS

01 Model domain & reference conditions

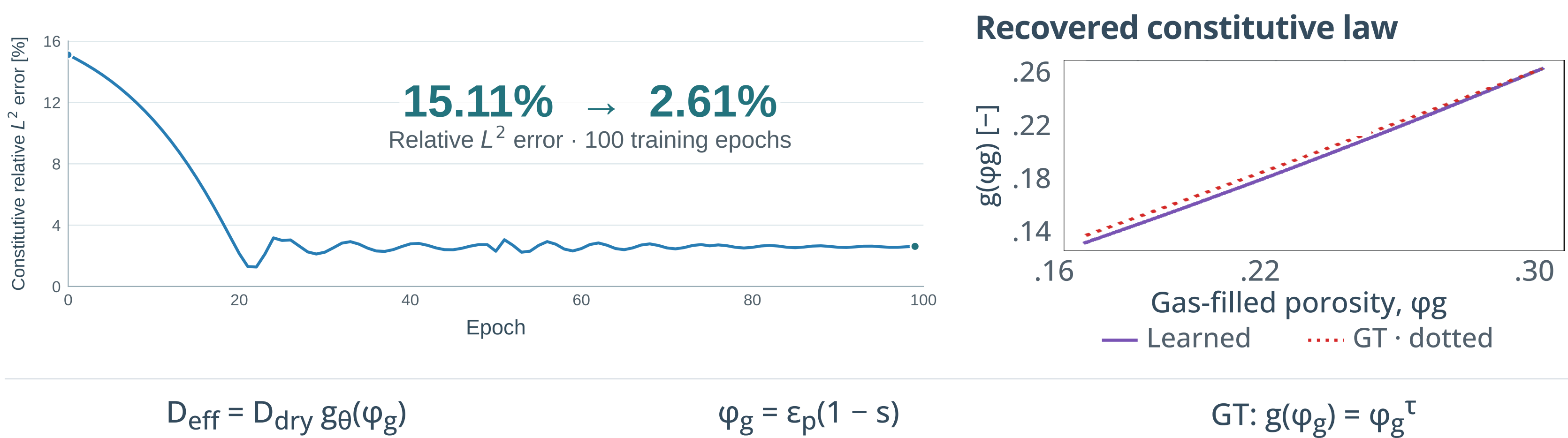


02 Water accumulation grows at high current



03 Recover the missing transport relation

Train at RH 0%: infer the gas-phase closure from synthetic FEM responses.



CONCLUSIONS & OUTLOOK

A learned transport law can be reused inside FEM without retraining.

01 Physically interpretable learning.

The recovered closure links gas-filled porosity to transport, making the learned relation directly inspectable.

02 Transfer within the tested conditions.

Agreement at changed RH and discretization supports reuse of the closure; it does not establish mesh convergence.

03 Next step: predictive validation.

Test against experimental responses and a wider operating range, then assess closure identifiability and uncertainty.

MODELING FRAMEWORK

PHYSICS-BASED FINITE-ELEMENT SOLVER

Known governing physics

- Flow through porous media
- Two-phase water transport
- Conservation equations retained

+

Learnable missing physics

Gas-phase transport closure

$$g_{\theta}(\varphi_g)$$

$$\varphi_g = \epsilon_p(1 - s)$$

MODEL OUTPUT
Predicted responses

Synthetic reference

COMPARE

$$L(\theta)$$

Response error

OPTIMIZER

Update θ only

Discrete-adjoint sensitivities

Known physics → FEM

Missing relation → ML

Only θ is trained

04 Zero-shot transfer with a frozen closure

